ATDL Assignment 2: Defensive Distillation

Total: 30 points

Objective: Investigate whether defensive distillation improves robustness against adversarial attacks on ResNet34/CIFAR-10 setup. Submit a runnable notebook with outputs and a short report containing plots, tables, and conclusions.

Experimental Setup

Use the CIFAR-10 training and test splits. Use ResNet34 architecture for all models.

Model	Training targets	Temperature
Baseline B	Ground-truth labels	1
Teacher M_1	Ground-truth labels	20
Student M_2	Teacher's soft targets	20

Question 1: Defensive Distillation (12 points)

(a) Freeze the teacher and train the student using

$$\mathcal{L} = -\frac{T^2}{N}\sum_{i=1}^{N}\sum_{k=1}^{10} q_{i,k}\log p_{i,k}, \qquad T = 20,$$

where $q_i = \text{softmax}(z_{M_1}(x_i)/T)$ and $p_i = \text{softmax}(z_{M_2}(x_i)/T)$. Explain the role of T^2. **[10 points]**

(b) Present training-loss curves and a table comparing clean accuracy and training time for all three models. **[2 points]**

Question 2: Adversarial Evaluation (18 points)

Use untargeted white-box attacks with true-label cross-entropy at $T = 1$. Apply perturbations in raw pixel space, enforcing

$$\|x_{\text{adv}} - x\|_\infty \le \epsilon, \qquad x_{\text{adv}} \in [0,1]^{3\times 32\times 32}.$$

Include any input normalization inside the attacked model.

(a) Implement FGSM and explain the role of ϵ. **[3 points]**

(b) Implement PGD with a random start, 20 iterations, and step size $\alpha = 2/255$. Explain its difference from FGSM. **[3 points]**

(c) Evaluate both attacks separately against each model at

$$\epsilon \in \{0, 2/255, 4/255, 8/255, 16/255\}.$$

Present an accuracy table and accuracy-versus-ϵ plots. Verify that $\epsilon = 0$ reproduces clean subset accuracy. **[4 points]**

(d) Show five original images alongside their FGSM and PGD versions against the student at $\epsilon = 8/255$. Include true and predicted labels. **[2 points]**

(e) Briefly discuss whether distillation improved robustness and whether smaller gradients imply robustness. **[6 points]**